The Midterm Penalty, 1858-2024

A very reliable rule, and what it still cannot tell you

84-355 Democracy's Data

August 2026

Knowing what an election will do is expensive. A survey of a thousand people costs real money and buys a few points of precision on one question in one state. A forecasting model costs a team. A campaign costs billions.

Against all of that, one sentence:

> **The president's party loses House seats at the midterm.**

That is the whole rule. No data collection, no model, no money. It has held for a century and a half. The question is what a rule that good is actually good for.

01 Direction is not what decides who governs

If the direction of a national legislative election is knowable in advance from a sentence, then a large part of what the midterm "means" was fixed before the campaign. Commentary that reads a midterm as a verdict on the president is reading a signal that the base rate already contained.

But there is a second reason, and it is the sharper one. **Control of the House is not decided by direction. It is decided by magnitude**, and by where the chamber happened to be sitting beforehand. A rule that predicts direction with near-certainty and magnitude with a spread of fifty seats is not a forecast of who governs.

02 A pattern this clean does not identify its own cause

Nobody disputes the pattern. The dispute is about the mechanism, and it has been running for sixty years without resolution.

- **Surge and decline.** Campbell (1960) proposed that presidential elections draw in peripheral voters who break toward the winner; at the midterm those voters stay home and the coattails withdraw.
- **Referendum.** Tufte (1975) modeled midterm outcomes as a function of presidential approval and the economy — voters punishing or rewarding the administration.
- **The presidential penalty.** Erikson (1988) tested the first two against the historical record, found both incompatible with it, and argued for a simple penalty: the midterm

electorate marks down the president's party for being the party in power, whatever it has done.
- **Balancing.** Bafumi, Erikson and Wlezien (2010) show that generic-ballot preferences drift toward the out-party across the midterm year, consistent with voters deliberately moderating policy.

Four live accounts, all consistent with the same table of numbers. Which of them is right cannot be settled by looking harder at the table, and the rest of this chapter does not try. It asks the prior question: what the table says at all.

03 The House counting itself, on a web page

The chamber's composition is not published by a statistical agency. It is published by the Office of the Historian of the U.S. House of Representatives. The page is called "Party Divisions of the House of Representatives, 1789 to Present," and it is institutional history rather than a statistical product. That is an unusual origin for a number this heavily used, and it is also the best one available. The House is the authority on how many of its own seats each party held, in a way no outside compiler can be. There is no rival file to check it against, and nothing here pretends otherwise.

It arrives as a web page. Not a spreadsheet, not an API, not a download. It is an HTML table meant for a person to read, which the build script has to fetch and take apart. The site turns away requests that do not look like a browser, so a fetch has to announce itself as one. That is the small, ordinary indignity of getting public numbers out of a government website, and it is worth noticing that it was necessary. The table's column layout also changes across eras as the party system does, so the script re-reads the header row whenever it meets one instead of assuming the Democrats are always in the same place.

Placed on the spectrum this book keeps returning to, the seat counts are as close to the source as anything in it: **made and published by the same body, and by the body being counted.** No researcher assembled them, no compiler reconciled them, nobody stands between the House and the number. But they were not *given*. They were **scraped**, off a page built for reading rather than for use, from a server that turns away anything not wearing a browser's name.

And the column this chapter actually turns on was **typed in by hand**. The party of the sitting president is not in the source, and was entered as twenty date ranges.

So the cost of getting this file is three things, none of them money. A disguise, to be let in at all. A repair, because the page's footnotes live inside the cells that hold the numbers. And one hand-keyed column with no upstream authority behind it, which is the only part of this chapter that could be wrong without anyone noticing.

The file is appropriate here because the unit is exactly right. The question is about the president's party gaining or losing House seats. A table whose row is a Congress, and whose columns are seats by party, answers it directly, with no stand-in anywhere in the chain. What would have been better is a table recording the chamber *on the day after each election*, and that is not what this is. Party divisions are given for the start of each Congress, and members die, resign and switch, which is why the source is footnoted so heavily.

Measuring change from one Congress's opening to the next is the standard convention, and it quietly credits the election with anything that happened in between.

Two of the difficulties are about measurement rather than about the file, and both are decisions somebody had to make. The first is that the two major parties do not account for the whole chamber. In 59 of these 84 elections some seats were held by neither, as many as 39 of them in 1858. So "the president's party lost seats" is a statement about one column of a table that does not add to its own total.

The second is larger, and it is disclosed here rather than buried: **the column naming the president's party is not in the source at all.** It was typed into the build script as a list of date ranges, from Buchanan to the present, and every claim in this chapter depends on it. Anyone can check it in a minute, and it is still the one column nobody downloaded.

The cleaning has one repair worth knowing about, because it is the kind that would never announce itself. The Historian's table footnotes many rows, and the footnote marker sits *inside* the cell with the number. Strip the HTML tags and the digits run together, so a 435-seat chamber carrying footnote 5 reads as `4355`. The script repairs any seat count above 999 by keeping its first three digits. That is a rule rather than a fact, and a reader rebuilding this file would have to apply it or notice the damage.

Everything else was subtraction. The series begins with the 36th Congress, on the judgment that Democratic-versus-Republican competition before that is not the same contest. The seat change for each election is the difference from the previous one, for whichever party held the White House at the time.

04 A web page is not a table

The Historian's page is one long HTML table, and the row for the Congress that the 2018 midterm elected looks like this inside it. Only the indentation has been added; every character is as the server sends it.

```
<tr>
  <td colspan="1" style="text-align: center;">
    <a href="/Congressional-Overview/Profiles/116th/"
       title="116th (2019-2021)">116th (2019-2021)</a></td>
  <td colspan="1" style="text-align: center;">435<sup>5</sup></td>
  <td colspan="1" style="text-align: center;">235</td>
  <td colspan="1" style="text-align: center;">199</td>
  <td colspan="1" style="text-align: center;">0</td>
  <td colspan="1" style="text-align: center;">5/1</td>
</tr>
```

To a person reading the page, that row says the chamber had 435 seats, that footnote 5 has something to say about it, and that the parties held 235 and 199. To a program, the seat cell contains `435<sup>5</sup>`. The ordinary way to turn a cell into a value is to throw away everything between angle brackets, and that turns it into this:

Congress	Seats	Democrats	Republicans	Other	Vacancies
116th (2019-2021)	**4355**	235	199	0	5/1

The chamber has just grown tenfold, and nothing anywhere has failed. No error is raised, no cell is empty, the row parses, the type is right. It is a plausible-looking integer in an integer column, and it would travel all the way to a chart.

The repair is one line of judgment: any seat count above 999 keeps its first three digits.

Quantity	Value
What a reader sees in the cell	435, then a raised 5
After the tags are thrown away	4355
After the repair	435
The largest the House has ever been	437

That rule is not a fact about the world. It is a bet on two of them: that the House will never seat a thousand members, and that the Historian's footnotes will not reach double digits. Both are true today. Neither is guaranteed, and the day either stops being true this file will go wrong quietly.

Here is the row that comes out:

Year	Congress	Seats	Dem	Rep	Pres. party	Midterm?	Seat change
2018	116	435	235	199	R	TRUE	-42

What had to be decided along the way

Which column is which changes as you scroll. The page is not one table with one header; it is a run of era-blocks, and a new header row appears whenever the party system changes. Early blocks label their columns *Pro-Administration* and *Anti-Administration*; later ones *Federalists* and *Democratic Republicans*, then *Adams* and *Jacksonians*, then *Democrats* and *Whigs*, and only after that *Democrats* and *Republicans*. Assume the Democrats are in the third column throughout, which is the obvious thing to assume, and the nineteenth century comes out as nonsense. The script instead re-reads the header every time it meets one and looks up the two columns by name.

Where the series starts is an argument, not a lookup. It begins at the 36th Congress, because before that the two columns are not the two parties this chapter is about. 84 elections survive that cut; everything before it is dropped, on a judgment about what counts as the same contest rather than on anything in the file.

The president's party is not in the file. Every claim in this chapter compares one party's seat count across two elections, and which party that is comes from a hand-typed list of date ranges. There is no column to join on and no source to check against — only the ordinary historical record, and a reader willing to spend a minute on it.

Change is measured between openings, not across an election. A row is a Congress as it was constituted on its first day, so the difference between two rows silently includes anything that happened in the fourteen months between the election and the swearing-in. That is the standard convention, and it is the reason the source is footnoted as heavily as it is. The footnote that started all this trouble is itself a note about members who arrived or left.

05 One row is one House election

Every number in this chapter comes out of one table, and one row of it looks like this.

Year	Congress	Seats in the chamber	Democrats	Republicans	President's party	Midterm?	Seat change for the president's party
2018	116	435	235	199	R	TRUE	-42

`seats` **is not a constant.** The chamber had 226 seats in 1866 and 435 today, and it has been as low as 183 and as high as 437. That variation turns out to matter more than it looks like it should.

A note on where the data came from. These counts come from the House's own party-division table, not from a roster of members. Rosters count *people*, and members who die or resign appear alongside their replacements, which is how a 435-seat chamber ends up "having" 449 members. **Seats and people are different units**, and choosing the wrong one is the kind of error that never announces itself.

06 Nine times in ten

Quantity	Value
Midterm elections in the data	41
President's party lost seats in	36
As a percentage	88%
Mean change	-32.5 seats
Median change	-28 seats

41 midterms since 1858, and the president's party lost seats in 36 of them — 88%.

Almost nothing in political science is this consistent, and it is worth stating plainly what it means: **the direction of a national election, nine times in ten, from a sentence you can say out loud.**

07 The midterms that went the other way

Five midterms went the other way, and they are worth knowing.

Year	President's party	Seat change	Chamber size
1866	R	37	226
1902	R	7	386
1934	D	9	435

Year	President's party	Seat change	Chamber size
1998	D	4	435
2002	R	8	435

The largest of them, 1866, is the one to be careful about. **1866 is not what it looks like:** Congress was refusing to seat delegations from the former Confederacy, so the chamber is not comparable to the one before it. Any pattern that includes it is partly measuring an argument about who counts as a state.

The other tail:

Year	President's party	Seat change	Chamber size
1894	D	-125	357
1874	R	-96	293
1890	R	-93	332
1922	R	-77	435
1938	D	-72	435

08 Trying to break it

If 1866 is a bad case, drop it and everything before 1878 with it.

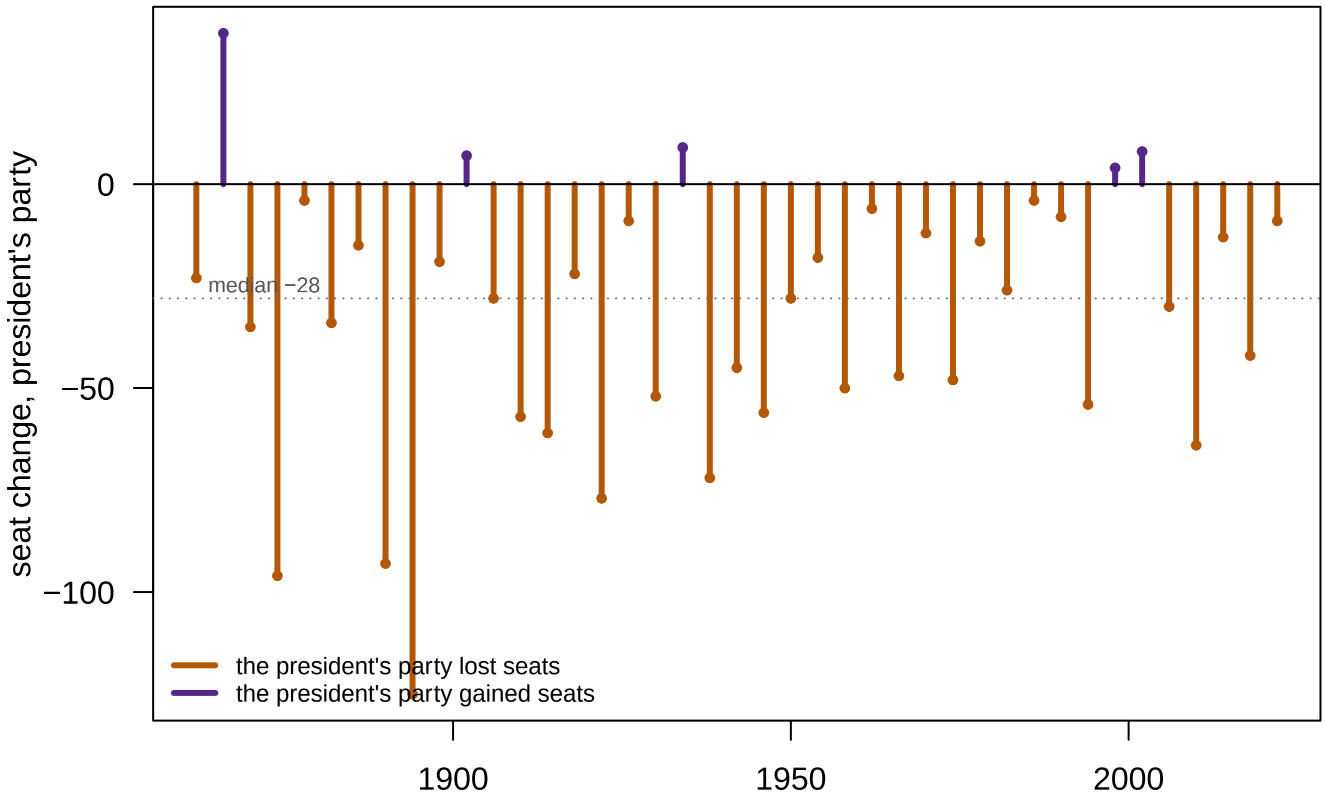
Sample	N	Lost seats	Mean change
All midterms	41	88%	-32.5
From 1878 onward	37	89%	-32.8

Almost nothing moves. That is informative rather than disappointing: one observation out of 41 cannot shift a mean far, and 1866 is a genuinely bad case on the merits *and* does not matter arithmetically. Those are two separate facts.

The corollary is worth noticing. **If no single bad case can move the result, then no single good case is holding it up either.** Robustness cuts both ways.

09 Every midterm since 1866

Robustness of that kind is easier to see than to state. Figure 1 puts every midterm on one axis.



Every midterm since 1866. The vertical axis is raw seats; the HTML version of this figure can also show the same change as a share of the chamber, which is the better unit for a chamber that changed size.

Figure 1. Every midterm since 1866, by the seat change for the president's party. One vertical line per election, colored by whether the president's party lost seats or gained them. The dotted line is the median, -28 seats.

The vertical axis here is raw seats. Chamber size is not constant across this figure, running from 183 to 437, and the same change is shown as a share of the chamber later in this chapter.

> **Before you read on, write a number down.** The direction is right 88% of the time and the median loss is 28 seats. **How many of the 41 midterms fall within ten seats either side of that median?** Commit to a count before you look.

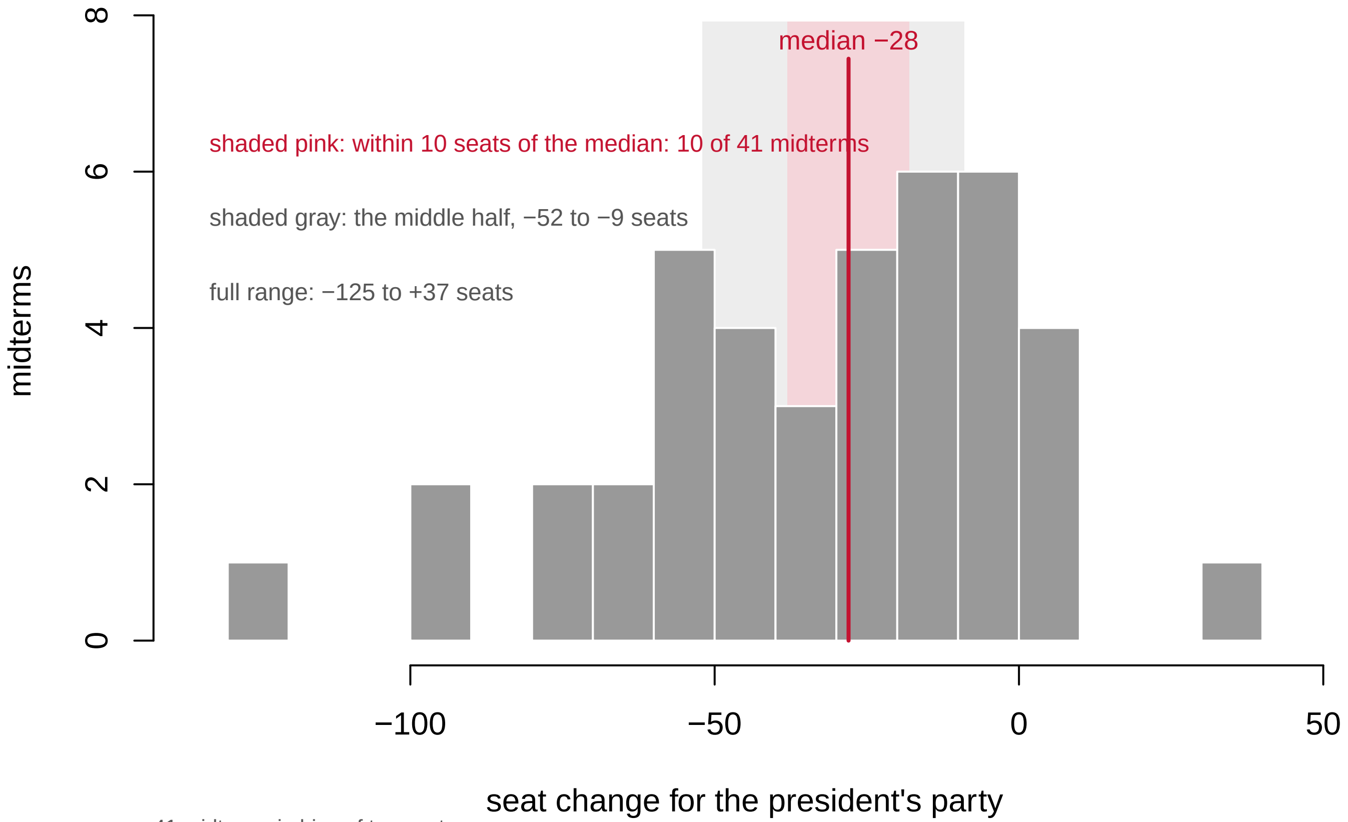
10 The sign is certain, the size is not

Statistic	Value
Median loss	-28 seats
Within 10 seats of the median	10 of 41 midterms (24%)
Middle half of outcomes (25th to 75th percentile)	-52 to -9 seats
Standard deviation	32.3 seats
Full range	-125 to +37 seats

Most readers write down a comfortable majority. A rule that gets the direction right nine times in ten sounds like a rule with a center, and a distribution with a center should put most of its mass near the middle.

Only 10 of 41 midterms, or 24%, land within ten seats of the median. The middle half of the distribution runs from a loss of 9 seats to a loss of 52.

A forecast that says "you will lose between 9 and 52 seats, probably, and possibly 125" is not much of a forecast. The rule is **nearly certain about the sign and close to uninformative about the size.**



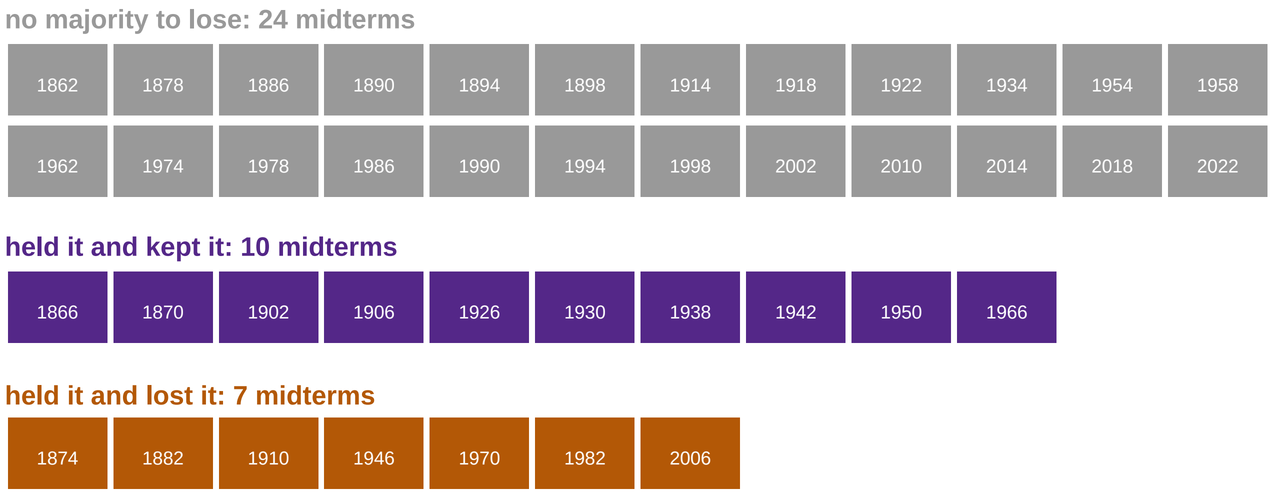
41 midterms in bins of ten seats.

Figure 2. 41 midterms, in bins of ten seats. The red line is the median, -28 seats; the pink band is ten seats either side of it and contains 10 of the 41 midterms. The gray band is the middle half of the distribution, -52 to -9 seats. The full range runs from -125 to +37.

And the size is what matters:

Quantity	Value
Midterms where the president's party held a House majority going in	17
... of those, midterms where it lost the majority	7
As a percentage	41%

The president's party entered 17 midterms holding a House majority and lost it in 7 of them — 41%. The direction of the seat change is close to a certainty; the consequence that anyone cares about is close to a coin flip. Those two sentences describe the same 41 elections.



One square per midterm. The president's party had a majority to lose in 17 of 41 and lost it in 7.

Figure 3. What happened to the majority. One square per midterm, grouped by whether the president's party had a House majority going in and whether it still had one coming out. It held a majority entering 17 of the 41 midterms and lost it in 7 of those — 41%. The same elections that make the direction near-certain make the consequence close to a coin flip.

11 Is the penalty shrinking? Per what?

The obvious next question is whether the rule is weakening. Split the midterms into eras and take the mean.

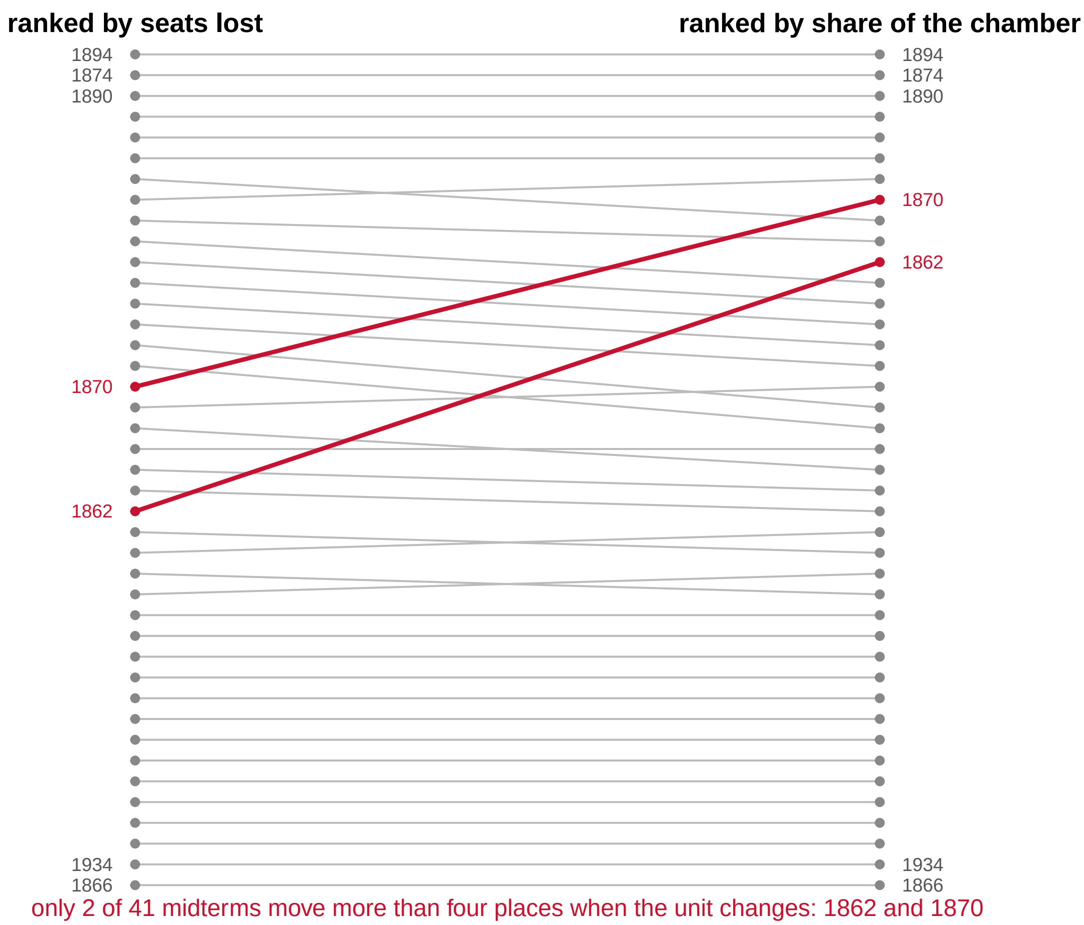
Era	Mean seat change	Mean % of the chamber	Mean chamber size	Midterms
1858-1900	-40.7	-12.8	294	10
1902-1950	-37.8	-8.8	425	13
1952-1980	-27.9	-6.4	435	7
1982-2000	-17.6	-4.0	435	5
2002-2024	-25.0	-5.7	435	6

Read the first column and the penalty looks halved: -40.7 in the nineteenth century against -25.0 since 2002.

But the chamber grew. It averaged 294 seats in the first era and 435 in the last. Losing forty seats out of three hundred is not the same event as losing forty out of 435, and the raw column compares them as though it were.

Normalizing does not rescue the "shrinking penalty" story — it **strengthens** it. The raw mean fell by 39% between the first era and the last; as a share of the chamber it fell by 55%. Both columns also show the same thing at the end: **the penalty falls through the twentieth century and rises again after 2002**, on either measure. "The penalty is shrinking" was never the right description.

Two warnings about that table. With 41 cases split five ways, some cells hold 5 elections, and almost any story can be produced by moving a boundary. And the individual cases reorder between units — 1862's loss of 23 seats is unremarkable in the first column and, out of a 184-seat chamber, 12.5% of the House.



Each line is one midterm, joining its rank by raw seats lost to its rank by share of the chamber.

Figure 4. The same midterms, ranked two ways. Each line joins one midterm's rank by raw seats lost to its rank by seats lost as a share of the chamber. Only 2 of 41 move more than four places when the unit changes, and they are 1862 and 1870. That is why the choice of unit looks harmless until it is not. The eras in the table above are built out of exactly this choice.

The habit worth taking from that figure is a question: **per what?** — asked before reading any trend, not after.

12 The six-year itch

Folk wisdom says a party's second midterm in power is worse than its first.

Group	N	Mean seats	Mean pct
Party held the White House 2 years or less	18	-35.6	-9.4
Party held the White House 6 years or more	11	-40.3	-11.3

The itch points the right way and the evidence is weak. The gap is about 5 seats on 11 observations, which is well inside what chance produces with groups this small — a two-sample test gives p = 0.69. The honest statement is that the folk wisdom is not contradicted and is not confirmed.

Note also that the definition did work here. "Second midterm" could mean the second midterm of one *president* or six years of one *party* holding the White House, and Truman, Johnson and Ford make those different. This uses party tenure. **The answer depends on a definition nobody states.**

13 Certainty about the sign, near-ignorance about the outcome

The president's party has lost House seats in 36 of 41 midterms since 1858; the median loss is 28 seats; and dropping Reconstruction changes neither. That much is as solid as anything in the study of American elections.

The sharper statement is the one that costs something. The rule is a direction, not a magnitude. Its interquartile range is 9 to 52 seats, and the president's party has lost its majority in 41% of the midterms where it had one to lose. **Certainty about the sign is compatible with near-ignorance about the outcome**, and the outcome is the part that decides who writes the laws.

Two things the table will not deliver, no matter how it is arranged. It will not say why the penalty happens: the four accounts above fit these numbers equally well, and nothing in a column of seat changes separates them. And it will not support "the penalty is shrinking," because on both measures it stopped shrinking after 2002 — and with 41 cases divided into five eras, era claims of any kind are fragile.

14 The mechanisms are still competing

Erikson (1988) is the sharpest statement of the problem. He tested withdrawn coattails, surge-and-decline and the referendum account against midterms since 1902 and found each incompatible with some feature of the record, proposing instead a flat penalty for holding the presidency. Bafumi, Erikson and Wlezien (2010) later showed that the drift toward the out-party happens *during* the midterm year and is visible in generic-ballot polling, which is evidence for deliberate balancing rather than mechanical turnout composition.

One of these accounts is testable here, and it fails. Suppose the penalty were mostly **regression to the mean**: the drift back toward average that follows an unusually good result, with nothing causing it but the arithmetic. Then midterms following large presidential-year gains should be worse. Across the 23 midterms where the file supplies a comparable preceding presidential-year change for the same party, the correlation is **-0.062**, essentially zero. That is a real result and a weak one. 23 cases cannot rule much out, and the comparison depends on the party holding the White House across both elections.

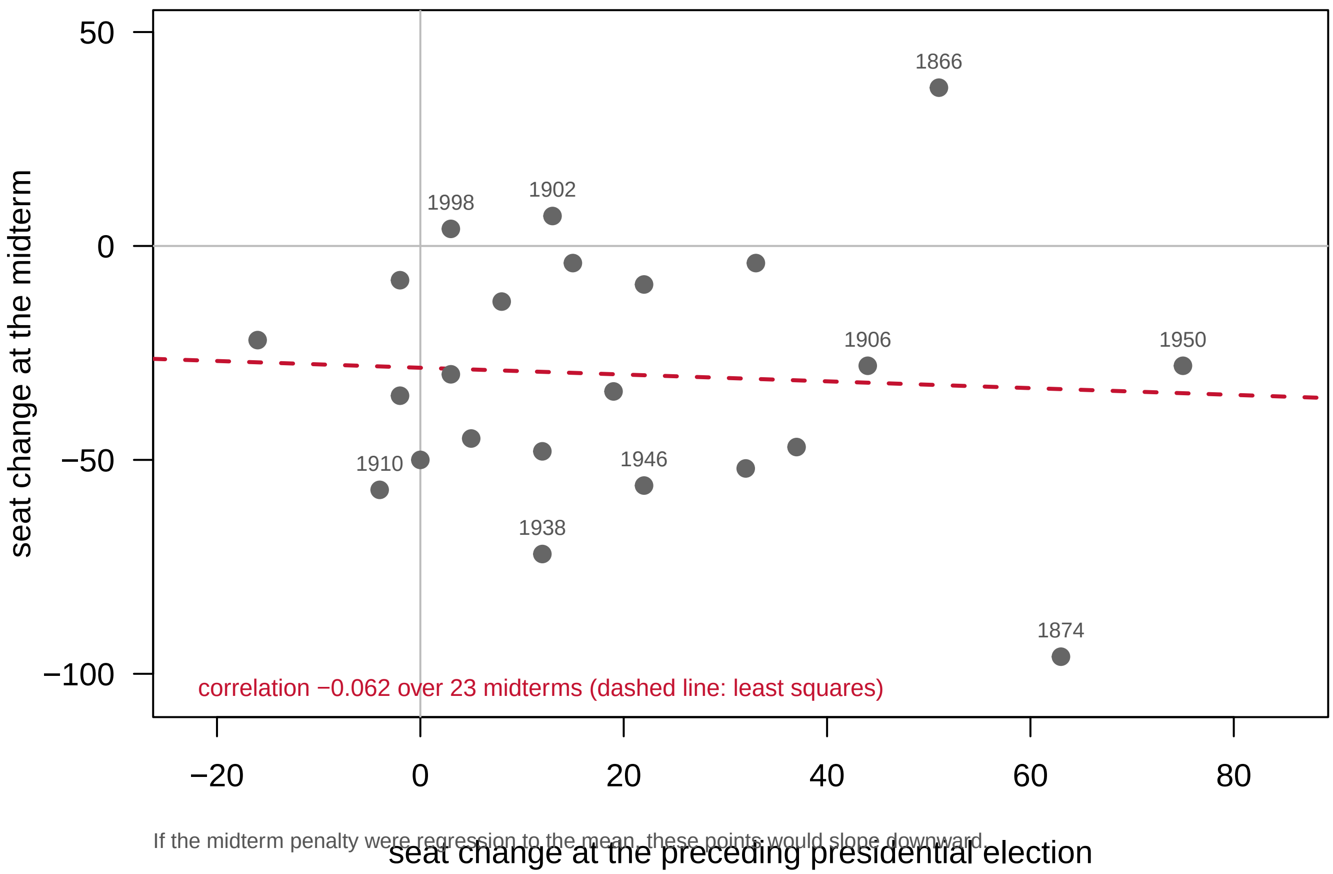


Figure 5. The midterm against the presidential election before it. One point per midterm for the 23 cases where the same party held the White House across both elections, placed by the seat change two years earlier and the seat change at the midterm. If the penalty were regression to the mean after presidential coattails, the cloud would slope downward. The dashed least-squares line is nearly flat; the correlation is -0.062.

The nationalization argument runs the other way. If congressional elections are increasingly votes about the president, as Hopkins (2018) and Abramowitz and Webster (2016) argue, the penalty should be getting *larger*, not smaller. The post-2002 rebound in the era table is the closest thing to a test here, and it is consistent with the nationalization claim while resting on 6 elections.

And there is a mechanical explanation for the long decline that has nothing to do with voters. Mayhew (1974) documented the disappearance of marginal House seats: as fewer districts sit near even, a given national swing moves fewer of them. A shrinking penalty may be a fact about the *map* rather than about the electorate: a claim about how districts are drawn rather than about how anyone votes. Nothing in a table of seat totals can tell those two apart.

15 What a century and a half of seat counts can testify to

The evidence behind all of this is a single consistent series: 84 House elections taken from the chamber's own party-division table, counted in the right unit. Seats, not members — a roster counts people, and members who die or resign appear alongside their replacements, which is how a 435-seat chamber ends up "having" 449 members. The series also records the size of the chamber in every year, which is what makes the "per what?" comparison possible at all; most published versions of this table do not carry it.

Its limits are specific rather than general.

One column, the party of the president at each election, is a hand-built lookup rather than something read out of a source. It is therefore the most likely place in the file for a mistake to be hiding. Seat change is not vote change. A seat swing depends on the map as much as on the electorate, and there is no vote column here at all, which is exactly why Mayhew's alternative cannot be tested.

41 cases is a small sample for any claim about a subgroup, and every subgroup claim above has been hedged for that reason. Starting in 1858 is a choice and not a discovery. Earlier Congresses sat under different party systems, and the source table's columns change names behind them. And the file contains nothing about the president, the economy, approval or the issues, so none of the four accounts of *why* the penalty exists can be tested against it.

That leaves three questions genuinely open. Which mechanism produces the penalty is the oldest of them: sixty years of work, four live accounts, no resolution, and a table that fits all four. Whether the post-2002 rebound is real or is 6 elections of noise cannot be settled until more midterms happen. And whether the long decline was ever about voters at all rather than about districts needs district-level data this file does not have — Mayhew's vanishing marginals are a testable alternative, not a settled one.

What can be said is the thing that was true on the first page and is more uncomfortable on the last. The most reliable regularity in American politics gets the direction of a national election right 88% of the time and tells you almost nothing about who ends up running the House.

16 Sources

Data

- Office of the Historian, U.S. House of Representatives, *Party Divisions of the House of Representatives, 1789 to Present* (https://history.house.gov/Institution/Party-Divisions/Party-Divisions/). The party of the president at each election is a hand-built lookup, typed in from the historical record.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`house_midterms.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `house_midterms.csv` has `pres_party_change` for every election, with a `midterm` flag. Compare midterms against the rest. How reliable is the rule, counted honestly?
- Find the midterms where the president's party gained seats. Read what happened in each of those years. Is there a pattern, or are they just exceptions?
- The rule tells you direction, not size. Sort the midterms by `pres_party_change` and look at the spread. Would knowing the direction have helped you predict who controls the House?
- `house_midterms.csv` records `seats` for each Congress, and the number changes. Find where the House grew or shrank, and work out what that does to a seat-change series.

Academic literature

- Abramowitz, A. I. and Webster, S. (2016). "The rise of negative partisanship and the nationalization of U.S. elections in the 21st century." *Electoral Studies* 41: 12–22.
- Bafumi, J., Erikson, R. S. and Wlezien, C. (2010). "Balancing, Generic Polls and Midterm Congressional Elections." *Journal of Politics* 72(3): 705–719.
- Campbell, A. (1960). "Surge and Decline: A Study of Electoral Change." *Public Opinion Quarterly* 24(3): 397–418.
- Erikson, R. S. (1988). "The Puzzle of Midterm Loss." *Journal of Politics* 50(4): 1011–1029.
- Hopkins, D. J. (2018). *The Increasingly United States.* Chicago: University of Chicago Press.
- Mayhew, D. R. (1974). "Congressional Elections: The Case of the Vanishing Marginals." *Polity* 6(3): 295–317.
- Tufte, E. R. (1975). "Determinants of the Outcomes of Midterm Congressional Elections." *American Political Science Review* 69(3): 812–826.

> REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT
>
> I want to rebuild a public dataset from its original source, without any starting files. Please do the following and show your work.

THE SOURCE. Office of the Historian, U.S. House of Representatives, ”Party Divisions of the House of Representatives, 1789 to Present”: https://history.house.gov/Institution/Party-Divisions/Party-Divisions/

One web page holding one table. Each row is a Congress, with the total number of seats and how many each party held. It is the House’s own count, and the authority on the question. No account or key is needed.

HOW TO FETCH IT. Two traps. The site refuses requests that do not identify themselves as a browser, so send a browser user-agent header. And the table’s footnote markers sit inside the cells: strip the HTML tags naively and 435 with footnote 5 comes out as ”4355”. Any seat count above 999 is a footnote stuck to a real number — keep the first three digits.

WHAT TO BUILD.

1. One row per House election from 1858 onward: the election year, the total seats, and the Democratic and Republican seat counts.
2. Mark which elections were midterms, work out which party held the presidency going into each one, and compute how many seats the president’s party gained or lost. Report the average midterm change.

CHECK YOUR WORK. The copy this book used was fetched in August 2026. If your rebuild matches it, you should find:

- 84 elections, 1858 through 2024
- the first row, 1858: 238 seats, 83 Democrats, 116 Republicans
- 2018: 235 Democrats, 199 Republicans — the president’s party down 42
- 435 seats in 55 of the 84 elections, and briefly 436 then 437 when Alaska and Hawaii joined ahead of the 1960s reapportionment

The Historian adds a row after each election and otherwise leaves the past alone; a new bottom row means time passed, not that you erred.